Venkata Sai Manas Reddy Naguru

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PROFESSIONAL SUMMARY

Aspiring Electrical and Electronics Engineer with a strong interest in Electric Vehicles, Powertrain Engineering, Power Systems, and Embedded Systems. Hands-on experience in microcontroller-based systems and sensor integration. Proficient in Python, C, and Embedded C, with exposure to MATLAB. Keen to apply core EEE knowledge and embedded firmware skills to real-world applications.

EDUCATION

Nitte Meenakshi Institute of Technology, Bengaluru 2024 - 2028
Bachelor of Engineering in Electrical and Electronics Engineering (CGPA: 8.33 / 10.0)

New Baldwin Independent PU College, Bengaluru 2023 - 2024
2nd PUC - 80.1%

PROJECTS

Predictive and Adaptive Synthetic Inertia Control (PASIC) for Renewable Power Systems

Technologies: Python, MATLAB/Simulink, TimeGAN | Timeline: Feb 2026 - May 2026
- Developed a TimeGAN-based generative AI engine to predict synthetic inertia control strategies for renewable grids.
- Architected a five-stage Sense -> Predict -> Compare -> Decide -> Actuate -> Log control pipeline.
- Integrated real-time frequency prediction with adaptive grid stabilization logic to suppress RoCoF excursions.

Smart Motor Health Monitoring and Predictive Maintenance System

Technologies: ESP32-S3, ACS712, MPU6050, LM35, I2C LCD, Hall Effect Sensor | Timeline: Mar 2026 - Apr 2026
- Designed an embedded system on ESP32-S3 to monitor industrial motor health in real-time.
- Implemented a three-tier health scoring algorithm to classify motor condition and trigger protective action.
- Integrated ACS712 current, LM35 temperature, MPU6050 vibration, and Hall Effect RPM sensors via I2C LCD.

RESEARCH AND ACHIEVEMENTS

- Authored the research paper "Predictive and Adaptive Synthetic Inertia Control for Renewable-Dominated Power Systems Using Generative AI", reaching the final submission stage of the IRAI Conference 2026.

TECHNICAL SKILLS

Languages: Python, C, MATLAB, Embedded C
Embedded & Hardware: ESP32, ESP32-S3, Arduino, Raspberry Pi, Sensor Fusion (Current, Temp, Vibration)
Engineering Tools: MATLAB/Simulink, Keil uVision, Code Composer Studio, PlatformIO, KiCad, Git/GitHub
Domains: Embedded Systems, Industrial Automation, IoT, EV Powertrain, Power Systems & Grid Stability

CERTIFICATIONS

- Introduction to Cybersecurity -- Cisco Networking Academy (Dec 2024)
- Getting Started with Artificial Intelligence -- IBM (Sep 2024)
- AI/ML for Geodata Analysis -- IIRS, ISRO (Sep 2024)
- Enterprise Design Thinking Co-Creator and Practitioner -- IBM (Jun 2024)
- Samsung Innovation Campus Certification (Coding and Programming) -- March 2025 to Present